

Operations Research — Fundamentals

Operations Research (OR) is the discipline of applying advanced analytical methods to help make better decisions. OR uses mathematical modeling, statistics, and optimization to arrive at optimal or near-optimal solutions.

Common OR techniques:

- Linear Programming (LP)
- Integer Programming
- Network Models (shortest path, max flow)
- Queuing Theory
- Simulation
- Decision Analysis

Linear Programming

Linear Programming (LP) is a mathematical technique for optimizing a linear objective function subject to linear equality and inequality constraints.

Standard form:

Maximize $c^T x$

Subject to $Ax \leq b, x \geq 0$

Key concepts:

- Decision Variables: quantities to be determined.
- Objective Function: the function to maximize or minimize.
- Constraints: limitations or requirements expressed as inequalities.
- Feasible Region: the set of all points satisfying constraints.
- Optimal Solution: the point in the feasible region that gives the best objective value.

The Simplex Method (Dantzig, 1947) is the classic algorithm for solving LP problems. It iterates through vertices of the feasible region.

Transportation and Assignment Problems

The Transportation Problem is a special type of LP concerned with shipping goods from multiple sources to multiple destinations at minimum cost.

Methods for finding an initial basic feasible solution:

- North-West Corner Method
- Least Cost Method
- Vogel's Approximation Method (VAM)

The Assignment Problem is a special case of the transportation problem where each source supplies exactly one unit and each destination demands exactly one unit. The Hungarian Method is the standard algorithm for solving assignment problems optimally.